

MAT 5023 01S

Problem-Solving Seminar



Course Syllabus Summer 2023

Course Information

Students will have the opportunity to engage in extensive experience and practice in solving mathematical problems.

Credit Hours: 3

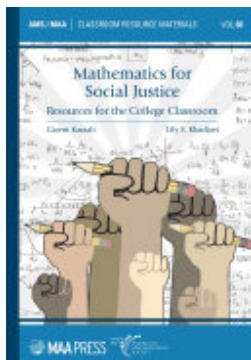
Course Modality: Traditional in-person

Learning Goals

At the end of the course, you will be able to:

The purpose of this course is to introduce you to the ideas teaching and learning mathematics through problem solving, with the objective of using mathematics to address real-world problems of social justice.

Course Materials



Mathematics for Social Justice: Resources for the College Classroom

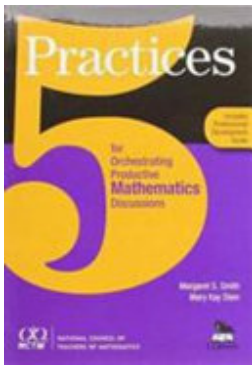
ISBN: 9781470449261

Authors: Gizem Karaali, Lily S. Khadjavi

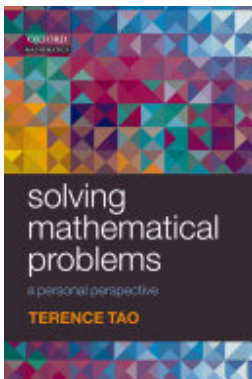
Publisher: American Mathematical Soc.

Publication Date: 2019-07-09

Required/Recommended/Optional: Required



Five Practices for Orchestrating Productive Mathematics Discussions
ISBN: 9781483351117
Authors: Margaret Schwan Smith, Mary Kay Stein
Publication Date: 2011-01-01
Required/Recommended/Optional: Required



Solving Mathematical Problems
ISBN: 9780191568695
Authors: Terence Tao
Publisher: OUP Oxford
Publication Date: 2006-07-28
Required/Recommended/Optional: Recommended

Activities and Grading

Activity	Quantity	%
Reading Responses	10	40
Problem Write-Ups	5	40
Final Project	1	20
Total		100%

Grade Distribution and Letter Grade

Letter Grade	Grade Range
A	84%-100%
B	75%-84%
C	65%-74%
D	50%-64%
F	less than 50%

Essential Student Information

- **Important:** Bookmark and visit the [Common Syllabus Information webpage](#) to find important and valuable resources about counseling services, transitory/minor medical issues, supplemental instruction, tutoring services, academic success coaching, sexual harassment and sexual misconduct, campus safety and emergency preparedness, inclusivity statement, and the Roadrunner Creed.
- For technical requirements, support, and academic resources, visit the [Student Support Gateway](#), where you can find all your tech and academic support resources in one place.
- Follow [Online Learning Netiquette](#) standards for your online communication activities. Please be mindful of the communication tools available in your course and use them for learning purposes. Class discussions take place in a respectful and safe environment, whether online or in person. UTSA encourages everyone to openly share their ideas and opinions without penalty or judgment, but learning should always be based on facts and research. It is possible to disagree without being disagreeable.
- UTSA provides reasonable accommodations to students via the [Student Disability Services](#). For more details on eligibility, policies, and requirements, please visit www.utsa.edu/disability or call (210) 458-4157.
- UTSA Wellbeing Resources: your wellbeing is a priority for us. UTSA is proud to partner with [Wellness 360](#) and [MySSP](#) to provide students with access to quality health and mental health care. Visit the [UTSA Students Wellbeing Resources](#) to explore the services available.

Instructor Contact Information

Instructor Name: Chris Duffer

Pronouns: they/them

Department

Mathematics

Office Location

FLN 4.01.30

Student Hours

MTWR 1:30 pm - 2:30 pm

or virtual meetings are available by appointment - please *email me* to schedule a time to meet.

Email Address: christopher.duffer@utsa.edu

Instructor Name: Priya V Prasad

Pronouns: she/her

Department

Mathematics

Office Location

FLN 4.01.52

Student Hours

MTWR: 1:30 pm - 2:30 pm

or virtual meetings are available by appointment - please use the [Calendly link here](#) to schedule a time to meet.

Email Address: priya.prasad@utsa.edu

Communication Plan

Dr. P: Emailing me is the best way to contact me. Short questions tangential to class may be answered on WhatsApp, but you may not get a timely answer.

Chris: Email is the best way to contact me, or you can talk to me before or after class

About Me

Dr. Prasad (she/her): Hi all! Y'all know me as your GAR (at least for the next few weeks!). My expertise and research is mathematics education, and I'm particularly interested in how teachers learn, process, and use mathematical and pedagogical knowledge to teach class. I got my PhD from the University of Arizona in 2014, and I have been at UTSA ever since I graduated.

Chris (they/them): Hello! I'm originally from San Antonio and originally came to UTSA for my undergraduate degree in math in 2015. I liked it here so much, I stayed to get my MS in mathematics education, and in 2021 I became a Lecturer in the math department. Currently I'm mostly teaching Calculus 1, but I'm super excited for the chance to work with you all this summer!

My Teaching Philosophy

This course is based on the idea that a powerful way to learn new mathematics is to engage in individual and collaborative problem-solving, revise and represent solutions, and connect thoughtfully to the thinking of others. As a required course in the MS for Mathematics Education degree program, the emphasis of this course is on both the experience of learning through problem-solving and the mechanics of teaching through problem-solving.

My Inclusivity Statement

The University of Texas at San Antonio, a Hispanic Serving Institution situated in a global city that has been a crossroads of peoples and cultures for centuries, values diversity and inclusion in all aspects of university life. As an institution expressly founded to advance the education of Mexican Americans and other underserved communities, our university is committed to ending generations of discrimination and inequity. UTSA, a premier public research university, fosters academic excellence through a community of dialogue, discovery, and innovation that embraces the uniqueness of each voice.

Assessments and Assignments

Reading Responses

There are 10 reading response assignments in this class (roughly 2 per week). All reading response assignments in this course will be graded on 2-1-0 scale. Responses that are complete, thorough, grammatically correct and well-edited, logically organized, and reflective will be given a grade of 2 points. Responses that do not meet this standard or are turned in late will be given 1 point. Failure to turn in anything by the end of the course will result in a 0. See the next page for more details.

Problem Write-Ups

You will be asked to write up your solutions to 5 problems that you worked on in class. These will also be graded on a 2-1-0 scale. Write-ups that are complete, well-explained, correct and logically organized will be given a grade of 2 points. Write-ups that do not meet this standard will be given 1 point and an opportunity for revision, if they are turned in on time. Late responses are accepted for the possibility of full credit, but are not eligible for revision. Failure to turn in anything by the end of the course will result in a 0.

Problem write-ups must be done digitally and submitted on Blackboard as a PDF. You may use LaTeX or a word processor such as Microsoft Word or Google Docs to create your problem write-up. You are strongly encouraged to create any appropriate visual representations digitally as well.

Final Project

You will complete a final project in groups that synthesizes ideas from this course and present it at the final meeting. The written portion of the project is a lesson plan for a problem-solving mathematics lesson that addresses a social injustice that is relevant to students at the high-school or college level. The presentation to your peers will include facilitating part of this lesson as you would in class.

Course Schedule

For a list of important university-wide dates, review [One Stop's academic calendar](#). This is a tentative schedule for the semester.

Date	Topic/Module	Assignments Due
July 6	Module 1: Problem Solving	
July 7	Module 1: Problem Solving	Reading Response 1
July 10	Module 1: Problem Solving	Problem Write-Up 1
July 11	Module 2: Voting	Reading Response 2
July 12	Module 2: Voting	
July 13	Module 2: Voting	Reading Response 3
July 14	Module 2: Voting	

Date	Topic/Module	Assignments Due
July 17	Interlude: Mondrian Puzzle	Problem Write-Up 2
July 18	Module 3: School Choice Problem	Reading Response 4
July 19	Module 3: School Choice Problem	
July 20	Module 3: School Choice Problem	Reading Response 5
July 21	Module 3: School Choice Problem	
July 24	Module 4: Geometry	Problem Write-Up 3
July 25	Module 4: Geometry	Reading Response 6
July 26	Module 4: Geometry	
July 27	Module 4: Geometry	Reading Response 7
July 28	Interlude: Cognitive Demand Activity	
July 31	Module 5:	Problem Write-Up 4
August 1	Module 5:	Reading Response 8
August 2	Module 5:	
August 3	Module 5:	Reading Response 9
August 4	Module 5:	
August 7	Work Day	Problem Write-Up 5
August 8	Presentations	Reading Response 10
August 9	Presentations	

Student Code of Conduct and Scholastic Dishonesty

The Student Code of Conduct is Section B of the Appendices in the Student Information Bulletin. Scholastic Dishonesty is listed in the Student Code of Conduct (Sec. B of the Appendices) under [Sec. 203](#).

Copyright and Fair Use

It is important to understand the issue of intellectual property rights. You may not use the images or thoughts of others for profit or gain without their written permission. The UTSA library has a [Copyright Laws and Public Performance Rights](#) (PPR) page.

Students with Disabilities

The University of Texas at San Antonio, in compliance with the Americans with Disabilities Act and Section 504 of the Rehabilitation Act, provides “reasonable accommodations” to students with disabilities. Only those students who have officially registered with Student Disability Services and requested accommodations for this course will be eligible for disability accommodations. Instructors at UTSA must be provided official notification of accommodation through Student Disability Services. Information regarding diagnostic criteria and policies for obtaining disability-based academic

accommodations can be found at www.utsa.edu/disability or by calling Student Disability Services at (210) 458-4157. Accommodations are not retroactive.

Family Educational Rights and Privacy Act (FERPA)

FERPA grants students the right to control certain disclosures of their educational records. For a full explanation of your rights and to grant access to FERPA educational records, go to [Student Catalog Annual FERPA Letter](#) and [One Stop Enrollment – FERPA Proxy Access](#). Without your consent or authorization of proxy access, UTSA may release Directory Information, such as but not limited to your name, email, phone, place of birth, and photograph, unless you have opted out of the release of [Directory Information](#). To opt out, go to [Restrict Directory Information Form](#).

Mandatory Reporting of Sexual Misconduct and Reporting of Health and Safety Information: If a student discloses an incident of sexual misconduct to any UTSA employee (other than to a designated confidential employee such as mental health counselor or PEACE advocate, a UTSA police officer using a pseudonym form or at a public awareness event), that information is not confidential, and the UTSA employee must report all known information to the UTSA Office of Equal Opportunity Services. Employees may also report any concerns about the health and safety of students or others to other school officials and/or law enforcement. For a complete list of exceptions to FERPA, please see [Student Catalog Annual FERPA Letter](#) and [HOP 5.01](#).

Standards for Reading Responses

Reading responses are three substantive paragraphs in which you will

1. summarize the article,
2. respond to the article from your own unique perspective, and
3. connect the article to the previous readings.

The following describes the dimensions upon which the reading responses will be evaluated. The point of these assignments is two-fold: 1) to allow you to reflect on the readings in relation to your own practice, and 2) to develop your communication skills about mathematics and education research.

These assignments will be graded on a 2-1-0 scale.

- **2 points:** Response meets all the standards for professionalism, explanatory thoroughness, clarity, and analysis shown below.
- **1 point:** Response meets all the standards for two of the three categories shown below.
- **0 points:** Response does not meet all the standards for at least two of the categories shown below.

Professionalism

- Responses should be typed and submitted under “Reading Responses” on Blackboard.
- Each paragraph should be written in an academic tone with the appropriate level of formality.

- You can still use first-person in academic writing (especially when responding to the articles from your own perspective), but these responses should not be written in conversational or informal style.
- Sentences should be complete, and distinct thoughts or themes should be organized into paragraphs.
- Responses should be edited before being posted.
 - There should not be more than 2 spelling or grammatical errors per 100 words.

Explanatory Thoroughness and Clarity

- You should generate at least 500 words for each response. You are always welcome to write more than this.
- Sentences should be clear and comprehensible. Each sentence should express a complete thought.

Level of Analysis

- Article summaries should show that you have read the article and have made an effort to think seriously about the main issues. Summaries should not be a restatement of the article's abstract, as the given abstract will not necessarily emphasize the true essence of the article. Reading Responses that cover multiple articles should include cogent summaries of both articles, as well as an analysis of the connections between the articles.
- Personal responses from your perspective should be brief and to the point – these sections should not take up more than a third of the total reading response. Each reading should be treated seriously, and the ideas should be engaged with from a respectful stance. Agreement or disagreement with the authors' assumptions or conclusions should be supported by facts, and critiques should reference particular aspects of the article.
- Connections to previous readings should be well-considered and thought out. These connections should be explained enough that your peers can follow your thought process for that connection.